

Microstructure and properties of AlCrFeCuNi_x ($0.6 \leq x \leq 1.4$) high-entropy alloys

Pi Jinhong^{a,b}, Pan Ye^{a,*}, Zhang Hui^{a,c}, Zhang Lu^a

^a School of Materials Science and Engineering, Southeast University, Jiangsu Key Laboratory of Advanced Metallic Materials, Nanjing 211189, China

^b School of Material Engineering, Nanjing Institute of Technology, Nanjing 211167, China

^c School of Material Science and Engineering, Anhui University of Technology, Maanshan 243002, China

ARTICLE INFO

Article history:

Received 23 July 2011

Received in revised form

18 November 2011

Accepted 19 November 2011

Available online 30 November 2011

Keywords:

Microstructure

Properties

High-entropy alloy

ABSTRACT

In this study, AlCrFeCuNi_x high-entropy alloys (HEAs) were prepared by vacuum arc melting and casting method in high-purity argon atmosphere. The microstructure and properties of as-cast alloys were investigated. The results showed that the content of nickel has a significant effect on microstructure and properties of AlCrFeCuNi_x HEAs. The as-cast AlCrFeCuNi_x ($0.6 \leq x \leq 1.4$) HEAs consisted of BCC phase, FCC phase, and one compound phase. The hardness of as-cast HEAs decreases obviously as x increases from 1.0 to 1.4. It is also found that, HEAs with equal molar components do not always show the best properties, while AlCrFeCuNi_{0.8} has the highest compressive strength and the best ductility among the tested HEAs.

© 2011 Elsevier B.V. All rights reserved.

1. Introduction

The conventional alloy systems are mainly based on one or two principal elements for primary properties. The addition of other minor elements is for acquiring definite microstructure and properties, such as alloy addition in steels, titanium-based alloys, and bulk metallic glasses (BMG). It is generally believed that intermetallic compounds tend to form in the alloys as the amount of alloying elements increases, which results in poor mechanical properties. Fortunately, this paradigm has been broken by high-entropy alloy systems developed by Yeh's research group [1]. High-entropy alloys (HEAs) are defined as alloys that contain at least five principal elements with each elemental molar ratio between 5 and 35 at.% [2,3]. Up to now, more and more researches demonstrated that the high-entropy alloys usually possess simple solid solution structures, and HEAs exhibit high strength, good ductility and resistances to wear, oxidation and corrosion [4–10].

Study on the hardness and microstructure of AlCrFeCuNi HEA has been reported by Li [11], and thermodynamic analysis of the simple microstructure was discussed. Li believes that the effect of high mixing entropy makes the Gibbs free energy of the solid solution AlCrFeCu smaller than that of inter-metallic compounds. Accordingly, the simple FCC and BCC solid-solution phases form prior to intermetallic compounds or more complicated phases. However, the 2θ scanning range of the X-ray diffract analysis was

limited between 20° and 60° . It is suggested that results with larger scanning range should be provided. On the other hand, since nickel is relatively high in cost, it is worthwhile to investigate whether addition of nickel could have great effects on microstructure and properties of the AlCrFeCuNi_x HEAs. In this paper, AlCrFeCuNi_x ($x = 0.6–1.4$) alloys were fabricated and the effects of nickel on the microstructure and properties were researched. It is aimed to provide a reference study for further physical investigations and basic data for potential application of these alloys.

2. Experimental details

Bulk AlCrFeCuNi_x HEAs with various content of nickel ($0.6 \leq x \leq 1.4$ in molar ratio) were prepared by a vacuum arc melting furnace under high-purity argon atmosphere. The ingots were obtained in a copper mould. The purity of bulk raw materials is higher than 99 wt%. After the alloys were melted with electromagnetic stir and cooled to solid subsequently, they were reversed and re-melted four times in order to assure chemical homogeneity. The samples cut from the solidified ingots were then heated to 650°C in a chamber electric furnace for 4 h to show the tempering resistance of the alloys. The microstructure and chemical composition of the alloys were analyzed by scanning electron microscope (SEM, JEOL JSM-6360LV) and energy dispersive spectrometry (EDS, GENESIS2000XMS60). The crystalline structures of the as-cast samples were identified by X-ray diffraction (XRD, D/Max2500 V) using Cu K α radiation. Vickers hardness measurement was conducted with a FM-700 hardness tester under a loading of 5 kg and a duration time of 15 s. Seven measurements were made on each sample

* Corresponding author. Tel.: +86 25 52090681.

E-mail address: panye@seu.edu.cn (P. Ye).

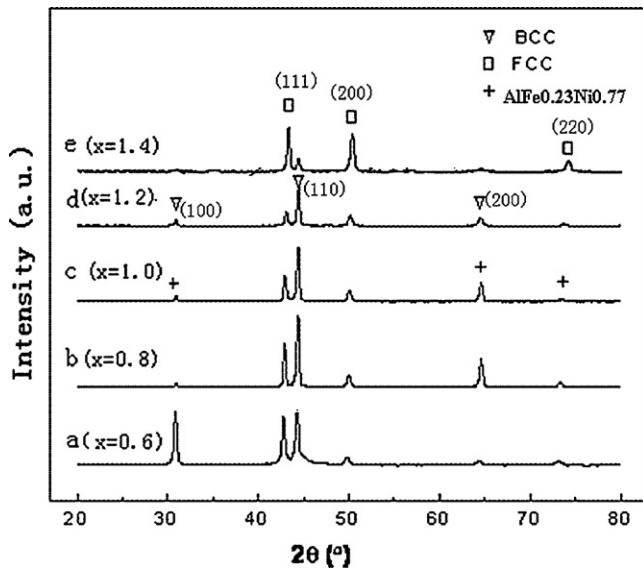


Fig. 1. XRD patterns of as-cast AlCrFeCuNi_x (0.6 ≤ x ≤ 1.4) high-entropy alloys.

for an average. Room-temperature compressive properties of the rod samples (φ 4 mm × 4 mm in size) were measured with an Instron-type testing machine (Instron MTS569) at a strain rate of $3 \times 10^4 \text{ s}^{-1}$.

3. Results and discussion

3.1. Microstructure characterization

X-ray diffraction (XRD) measurements were employed to investigate the phase and structure of the as-synthesized samples. As shown in Fig. 1, when the value of x changes between 0.6 and 1.4, all HEAs are composed of three phases, which are BCC, FCC and a compound phase AlFe_{0.23}Ni_{0.77} (the JCPDS file number is 47-1126). Obviously, the microstructures of these tested HEAs are relatively simple compared with traditional alloys with so many component elements. Fig. 2 shows the metallographic images of the as cast AlCrFeCuNi_x HEAs. From Fig. 2, there are no remarkable differences among the microstructure of the tested HEAs, except for slightly changes occurred in the proportion of phases and grain sizes with different x . Combine Figs. 1 and 2, it can be easily seen that nickel does have impact on the microstructure of the tested HEAs. As x changes, the ratio of three phases changes, too.

According to the Gibbs phase rule, $f = c - p + 1$ (f : degree of freedom, c : number of elements, and p : number of existing phase), the maximum number of equilibrium phases in a c component system is $c + 1$. And $p > c + 1$ when solidified in non-equilibrium conditions. Compared with conventional alloys, the number of phases in the tested HEAs is well below the maximum number calculated by the Gibbs phase rule.

As we know, the value of mixing entropy (ΔS^{mix}) of HEAs is higher than that of most inter-metallic compounds, leading to the preferential formation of solid solutions. Besides, the formation of solid solutions and compound may be affected by many other factors such as chemical bond, dimension of atom, electron concentration, difference of electro-negativity, and so on. Among which the difference of electro-negativity and dimension of atoms play an important role. The value of atomic radius and electro-negativity [13–15] of component elements in this paper are shown in Table 1.

From Table 1, the aluminum element and Fe/or Ni has a maximum difference of atomic radius and small difference of electro-negativity value. The larger differences of atomic radius

Table 1

The value of atomic radius and electro-negativity.

	Al	Cr	Fe	Cu	Ni
Atomic radius (nm)	0.143	0.125	0.124	0.127	0.124
Electro-negativity	1.90	1.66	1.83	1.90	1.91

Table 2

Chemical composition of as-cast AlCrFeCuNi_{1.0} in atomic percentage by EDS.

	Al	Cr	Fe	Cu	Ni
Dark phase	8.7	39.77	35.41	5.13	10.99
White phase	15.25	2.72	6.03	62.37	13.64

results in larger interstice for atoms diffuse which in favor of the formation of compounds. The smaller difference of electro-negativity value also makes easier formation of compound. That is why only small amount AlFe_{0.23}Ni_{0.77} was detected, no any other inter-metallic compounds was found in the as cast AlCrFeCuNi_x (0.6 ≤ x ≤ 1.4) HEAs.

In order to find out the differences among the phases, AlCrFeCuNi_{1.0} HEA was analyzed by EDS. The chemical composition of as-cast AlCrFeCuNi_{1.0} is showed in Table 2. From Table 2, Cu segregates seriously, which should be attributed to the small bonding energies of Cu with other atoms [12]. As mentioned above, the tested HEAs are composed of BCC, FCC and AlFe_{0.23}Ni_{0.77} phase. So, it can be concluded that the CrFe-rich dark phases and the Cu-rich white phases are two different solid solution phases. Considering the fact that Cu exhibits a FCC structure, whereas the pure Cr and Fe show BCC structure, and Cu with a FCC structure promotes the formation of a FCC solid solution [15], the Cu-rich phase and CrFe-rich phase can be identified as FCC phase and BCC phase, respectively.

For better understanding the volume fraction of each phase on the properties of the tested alloys, the volume fractions of each phase are roughly calculated by the following formula [16].

$$w_{p_i} = \frac{p}{\sum p_i} \times 100\%$$

where w_{p_i} is the volume fraction of the i th phase, p is the peak intensity of a given phase in the XRD pattern, i is the number of the phase, and $\sum p_i$ is the total peak intensity of all phases in the XRD pattern. The volume fractions of each phase are presented in Table 3. In particular, it is disfavoring the potential forming tendency of AlFe_{0.23}Ni_{0.77} phase when x deviating from the value of 1.0.

3.2. Hardness

The hardness of the as cast AlCrFeCuNi_x (0.6 ≤ x ≤ 1.4) HEAs are showed in Fig. 3. It can be clearly seen that the tested alloys have relatively high hardness, especially when the value of x is less than 1.0. The high hardness may result from the highly distorted crystals with multi elements which result in strong solid solution strengthening.

As x increases from 1.0 to 1.4, the hardness of HEAs decreases markedly. According to Table 3, the proportion of CrFe-rich BCC phase decreases with x (1.0 ≤ x ≤ 1.4). The results shows that the

Table 3

The volume fraction of each phase for each tested alloy (%).

x	BCC phase	FCC phase	AlFe _{0.23} Ni _{0.77}
0.6	51.0	43.4	5.6
0.8	50.7	30.1	19.2
1.0	54.9	26.7	18.4
1.2	48.5	28.7	22.8
1.4	44.6	51.6	3.8

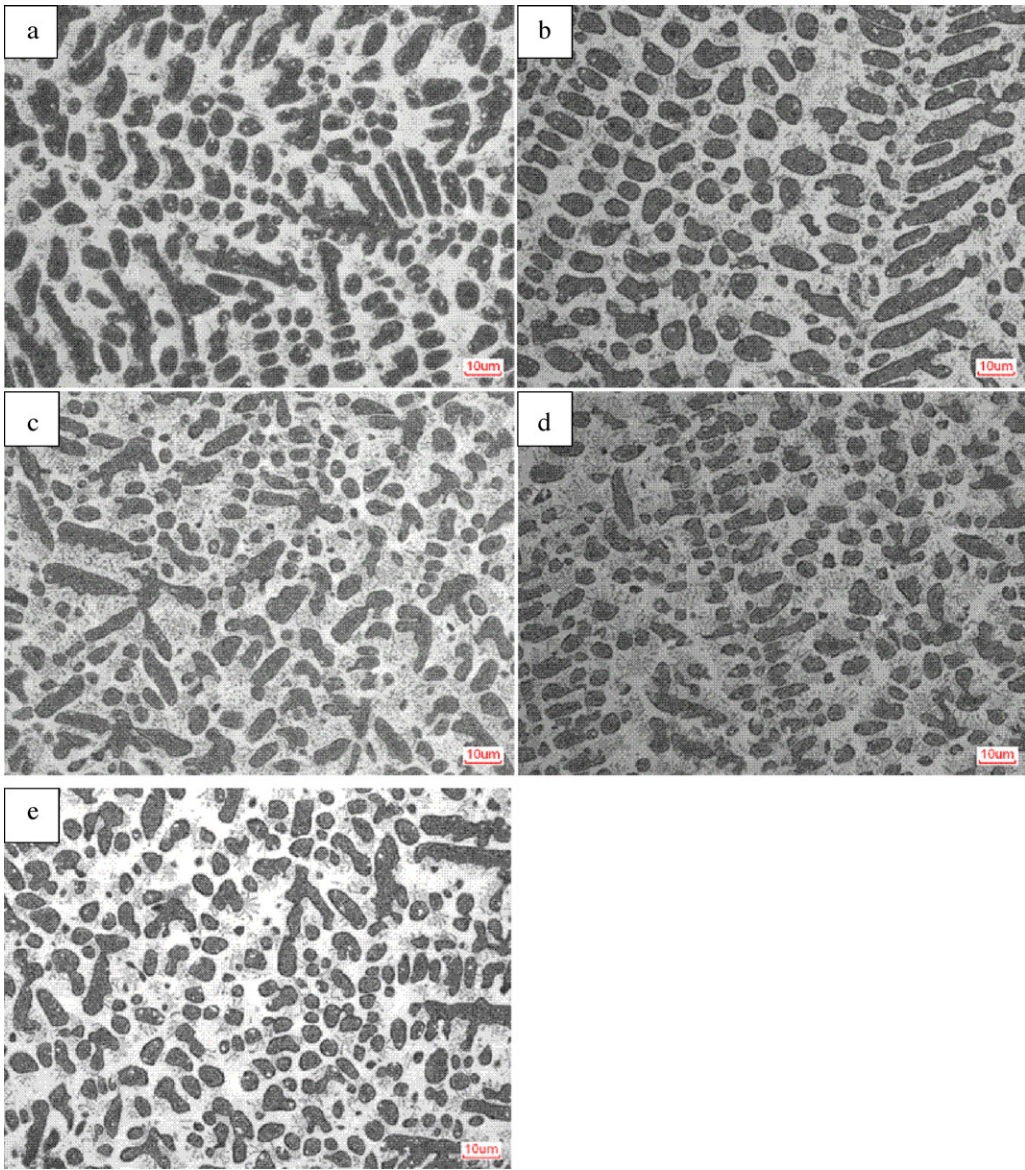


Fig. 2. Metallographic images of as-cast AlCrFeCuNi_x (x from 0.6 to 1.4) high-entropy alloys: (a) x = 0.6; (b) x = 0.8; (c) x = 1.0; (d) x = 1.2; (e) x = 1.4.

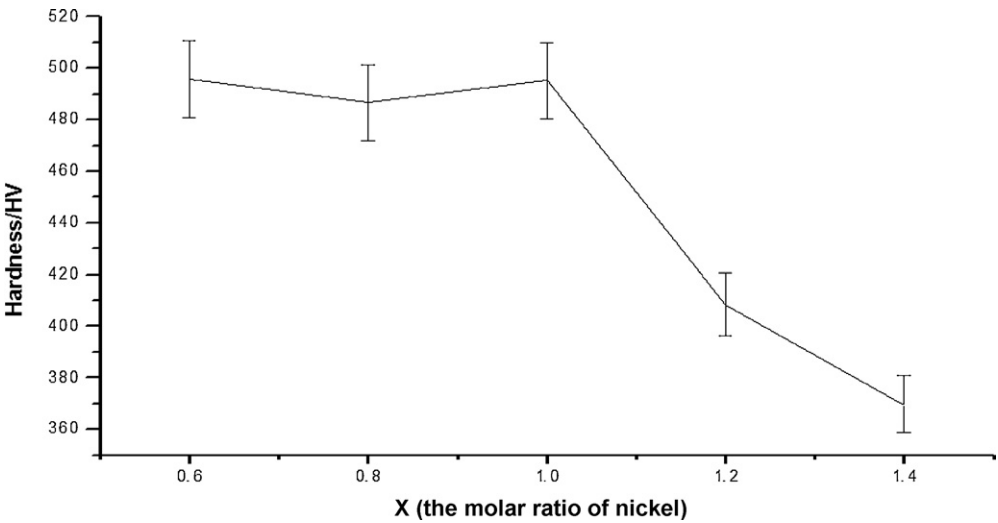


Fig. 3. Hardness of the as cast AlCrFeCuNi_x (x from 0.6 to 1.4) high-entropy alloys.

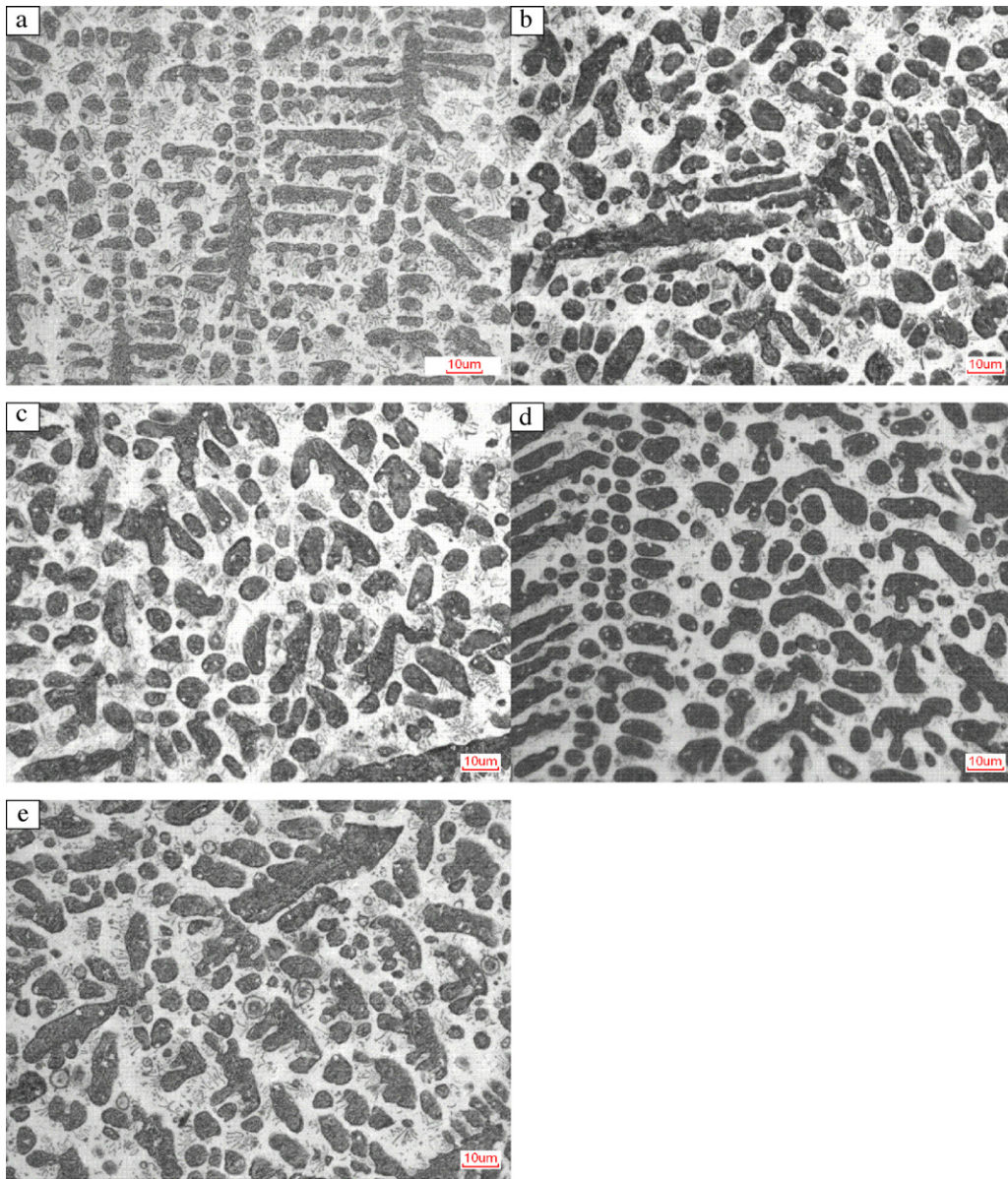


Fig. 4. Microstructures AlCrFeCuNi_x HEAs after treated at 650 °C for 4 h: (a) $x = 0.6$; (b) $x = 0.8$; (c) $x = 1.0$; (d) $x = 1.2$; (e) $x = 1.4$.

CrFe-rich BCC phase contributes more to the hardness of HEAs than Cu-rich FCC phase does. The reason may be contributed to fact that the closest packing planes in BCC structure are denser and possess a smaller inter-planar spacing and higher lattice friction for dislocation motion than that of FCC structure. AlCrFeCuNi_{0.8} is slightly softer than AlCrFeCuNi_{0.6} and AlCrFeCuNi_{1.0}. Certainly, the difference of hardness between AlCrFeCuNi_{0.8} and AlCrFeCuNi_{0.6} is very slight considering the measuring error. And this abnormal phenomenon may be caused by the relatively less sharp-pointed grains of AlCrFeCuNi_{0.8}, which results in less crystal distortion and less strengthening.

3.3. Tempering resistance

Fig. 4 shows the variation in microstructure of heat treated AlCrFeCuNi_x HEAs. It can be seen that the microstructures do not coarsen obviously for 4 h under 650 °C, which means the tested alloys are tempering resistant. The interaction between the coexisting multi elements restricts the atoms' diffusion, leading to the stabilizing of microstructure accordingly.

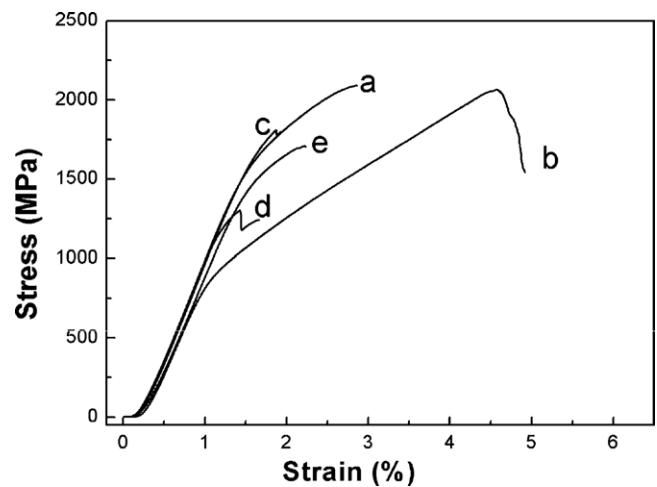


Fig. 5. Stress-strain curves of AlCrFeCuNi_x (x from 0.6 to 1.4) high-entropy alloys under compression: (a) $x = 0.6$; (b) $x = 0.8$; (c) $x = 1.0$; (d) $x = 1.2$; (e) $x = 1.4$.

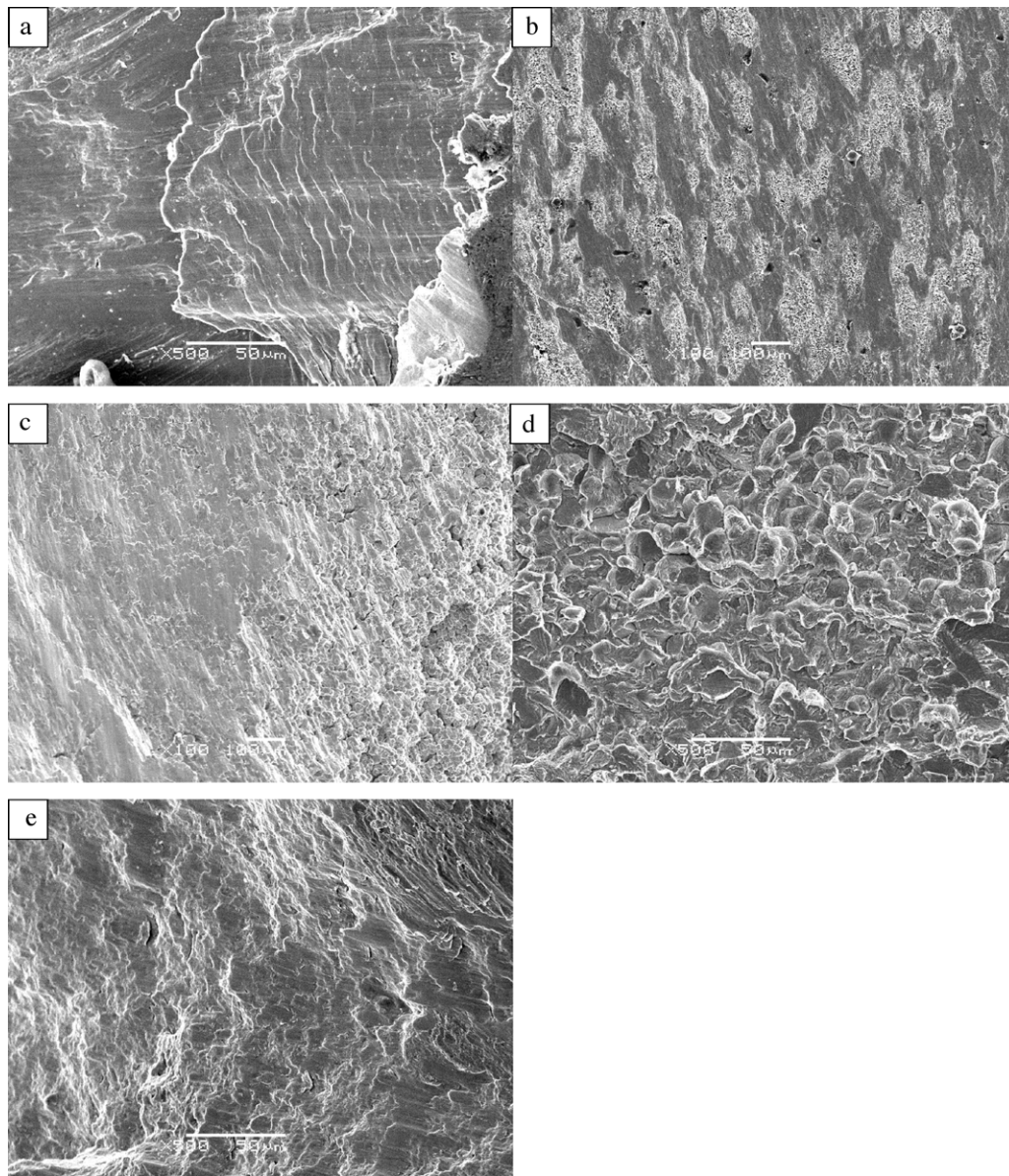


Fig. 6. Fractographic features of AlCrFeCuNi_x (x from 0.6 to 1.4) HEAs: (a) $x=0.6$; (b) $x=0.8$; (c) $x=1.0$; (d) $x=1.2$; (e) $x=1.4$.

With a view of most steels cannot stabilize at such high temperature, AlCrFeCuNi_x HEAs show good ability of tempering resistance. Hence these HEAs have advantages when used at high temperature. The results also show that higher nickel content does not assure better tempering resistance for AlCrFeCuNi_x HEAs.

3.4. Compressive experiment

Fig. 5 shows the compressive curves of AlCrFeCuNi_x (x from 0.6 to 1.4) HEAs at room temperature. The corresponding fractographic features were shown in Fig. 6.

As it can be seen, BCC phase contributes more to the compressive strength than FCC phase does. AlCrFeCuNi_{1.2} and AlCrFeCuNi_{1.4} which consisted of less proportion of BCC phases possess the lowest compressive strength among the tested HEAs. Meanwhile, too much AlFe_{0.23}Ni_{0.77} phase may result in stress concentration which is unfavorable to the strength. The fracture surface of AlCrFeCuNi_{0.6}, AlCrFeCuNi_{0.8} and AlCrFeCuNi_{1.0} reveals deformed and sheared regions. The alloy AlCrFeCuNi_{0.8} corresponds to the most heavily deformation and shear band. It indicates that the fracture

mode of AlCrFeCuNi_{0.8} is inter-crystalline fracture accompanied by ductile rupture. The fracture surface of AlCrFeCuNi_{1.2} consists of quasi-cleavage facets, while AlCrFeCuNi_{1.4} consists of dimple-like structures. Therefore, too much nickel is not beneficial to improving the compressive strength, and we do not suggest x more than 0.8 in order to reduce the cost.

4. Conclusions

- (1) The AlCrFeCuNi_x (x from 0.6 to 1.4) HEAs are composed of BCC phase, FCC phase and AlFe_{0.23}Ni_{0.77} phase, respectively.
- (2) The hardness of alloys decreases obviously as x increases from 1.0 to 1.4.
- (3) AlCrFeCuNi_{0.8} has the highest compressive strength and the best ductility among the tested HEAs.

References

- [1] J.W. Yeh, S.K. Chen, S.J. Lin, et al., J. Adv. Eng. Mater. 5 (2004) 299–303.
- [2] T.T. Shun, Y.C. Du, J. Alloys Compd. 478 (2009) 269–272.

- [3] V. Dolique, A.L. Thomann, P. Brault, et al., *J. Surf. Coat. Technol.* 204 (2010) 1989–1992.
- [4] F.J. Wang, Y. Zhang, G.L. Chen, *J. Alloys Compd.* 478 (2009) 321–324.
- [5] V. Dolique, A.L. Thomann, P. Brault, et al., *J. Mater. Chem. Phys.* 117 (2009) 142–147.
- [6] C.W. Tsai, Y.L. Chen, M.H. Tsai, et al., *J. Alloys Compd.* 486 (2009) 427–435.
- [7] C.Y. Hsu, T.S. Shen, J.W. Yeh, et al., *J. Wear* 268 (2010) 653–659.
- [8] Y.Y. Chen, T. Duval, U.D. Hung, et al., *J. Corros. Sci.* 47 (2005) 2257–2279.
- [9] Y.J. Zhou, Y. Zhang, Y.L. Wang, et al., *J. Mater. Sci. Eng. A* 454–455 (2007) 260–264.
- [10] Y.S. Huang, L. Chen, H.W. Lui, et al., *J. Mater. Sci. Eng. A* 457 (2007) 77–83.
- [11] A.M. Li, X.Y. Zhang, *J. Acta Metall. Sin. (Engl. Lett.)* 22 (2009) 219–224.
- [12] Y.P. Wang, B.S. Li, M.X. Ren, et al., *J. Mater. Sci. Eng. A* 491 (2008) 154–158.
- [13] B.W. Zhang, W.Y. Hu, X.L. Shu, *Theory of Embedded Atom Method and Its Application to Materials Science—Atomic Scale Materials Design Theory*, ChangSha, Hunan University publishing company, 2002.
- [14] W.F. Gale, T.C. Totemeier, *Smithells Metals Reference Book*, Amsterdam, Elsevier, 2004.
- [15] C. Li, J.C. Li, M. Zhao, et al., *J. Alloys Compd.* 475 (2009) 752–757.
- [16] B. Ren, Z.X. Liu, B. Cai, et al., *J. Mater. Des.* 33 (2011) 121–126.